

TEST DRIVEN DEVELOPMENT - MOCKITO

ExternalApi.java:

```
package com.prerna.mockito;

public interface ExternalApi {
    String getData();
}
```

MyService.java:

```
package com.prerna.mockito;

public class MyService {
    private ExternalApi externalApi;
    public MyService(ExternalApi externalApi) {
        this.externalApi = externalApi;
    }
    public String fetchData() {
        return externalApi.getData();
    }
}
```

MyServiceTest.java:

```
package com.prerna.mockito;

import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.assertEquals;
import static org.mockito.Mockito.*;

public class MyServiceTest {

    @Test
    //Exercise 1: Mocking and Stubbing
    void testExternalApi() {
```

```
ExternalApi mockApi = mock(ExternalApi.class);

when(mockApi.getData()).thenReturn("Mock Data");

MyService service = new MyService(mockApi);

String result = service.fetchData();

assertEquals("Mock Data", result);

}
```

@Test

//Exercise 2: Verifying Interaction

```
void testVerifyInteraction() {

    ExternalApi mockApi = mock(ExternalApi.class);

    MyService service = new MyService(mockApi);

    service.fetchData();

    verify(mockApi).getData();

}
}
```

TestDrivenDev

Version control

MyServiceTest

Project

exercise4

Calc

mockito

ExternalApi

MyService

resources

test

java

com.perna

exercise2

exercise3

exercise4

mockito

MyServiceTest

HelloTest

target

.gitignore

pom.xml

1

2

3

4

5

6

7

8

9

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20

```
package com.perna.mockito;

import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.assertEquals;
import static org.mockito.Mockito.*;

public class MyServiceTest {

    @Test
    void testExternalApi() {

        ExternalApi mockApi = mock(ExternalApi.class);

        when(mockApi.getData()).thenReturn("Mock Data");

        MyService service = new MyService(mockApi);

        String result = service.fetchData();
```

Run

MyServiceTest

MyServiceTest (com.perna) 1 sec 681 ms

testExternalApi() 1 sec 666 ms

testVerifyInteraction() 15 ms

2 tests passed

2 tests total, 1 sec 681 ms

```
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
WARNING: A Java agent has been loaded dynamically (C:\Users\Perna Premkumar\.m2\repository\
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes
```